# ESTIMATION THEORY

**Total Marks: 20**
**Show complete working of each mcqs**
**All Questions are Compulsory**

**Q1.** A random sample of size 12 from Bernoulli($p$) contains 8 successes. The MLE of $p$ is:

A) 0.50

B) 0.67

C) 0.75

D) 0.80

**Q2.** Using the result of Q1, the variance of the MLE $\hat{p}$ is:

A) 0.018

B) 0.222

C) 0.015

D) 0.25

**Q3.** Let $X_1, X_2, X_3$ be independent random variables with $E(X_i) = \mu$ and $Var(X_i) = 9$. Consider the estimator:

$$\hat{\mu} = \frac{X_1 + X_2 + X_3}{3}$$

Variance of the estimator equals:

A) 3

B) 9

C) 1

D) 27

**Q4.** Suppose

$$\hat{\mu} = \frac{2X_1 + X_2}{3}$$

and $E(X_1) = E(X_2) = \mu$.

Bias of the estimator is:

A) 0

B) $\mu/3$

C) $-\mu/3$

D) $\mu$

**Q5.** If $Var(X_1) = 4$ and $Var(X_2) = 9$ (independent),
Variance of

$$\hat{\mu} = \frac{2X_1 + X_2}{3}$$

is:

A) 2

B) 2.78

C) 1.78

D) 3

**Q6.** Suppose $X_1, X_2$ are independent with $E(X_1) = E(X_2) = \mu$, $Var(X_1) = 10$, $Var(X_2) = 15$.
Consider

$$\hat{\mu}_1 = \frac{X_1 + X_2}{2}$$

and

$$\hat{\mu}_2 = \frac{3X_1 + X_2}{4}$$

Which estimator has smaller variance?

A) $\hat{\mu}_1$

B) $\hat{\mu}_2$

C) Equal

D) Cannot determine

**Q7.** Let $X_1, \ldots, X_4$ be a sample from $N(\mu, 16)$.
Sample values: 10, 14, 12, 8.
The MLE of $\mu$ equals:

A) 11

B) 12

C) 10

D) 11.5

**Q8.** For Q7, variance of the MLE equals:

A) 4

B) 16

C) 2

D) 8

**Q9.** Suppose

$$\hat{\theta} = X + 5$$

where $E(X) = 3$, $Var(X) = 6$, and true $\theta = 3$.

The MSE equals:

A) 6

B) 31

C) 25

D) 11

**Q10.** If an estimator is unbiased and variance = 2.5, then MSE equals:

A) 0

B) 2.5

C) 5

D) Cannot determine